



UNIVERSITY^{AT}ALBANY
STATE UNIVERSITY OF NEW YORK

CSI 436/536 (Spring 2026)

Machine Learning

Lecture 13: Error Decomposition

Chong Liu

Department of Computer Science

Mar 30, 2026

Recap: Estimating mean of Gaussian distr.

- Data

$$X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \mathcal{N}(\mu, \sigma^2)$$

- Likelihood:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x - \mu}{\sigma}\right)^2\right)$$

- The MLE problem:

$$\hat{\mu} = \arg \max_{\mu \in [0,1]} \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(X_i - \mu)^2}{2\sigma^2}}$$

Recap: MLE for linear regression

- $P(y|x)$ is modeled by “Linear Gaussian model”

$$y_i = x_i^T \theta^* + \epsilon_i \quad \text{where } \epsilon_i \sim \mathcal{N}(0, \sigma^2) \text{ i.i.d.}$$

- Data:

$$(x_1, y_1), \dots, (x_n, y_n)$$

- In-class exercise: Work out the optimization problem to solve for the MLE for θ^* .
 - Solution: Same as linear regression using Director Solver.

After we fit the MLE, how to make predictions?

The idea is to just “Plug-In”

- For classification problems

$$h^*(x) = \max_y p_{\theta}(y|x) \xrightarrow{\text{Plug in}} \hat{h}(x) = \max_y p_{\hat{\theta}}(y|x)$$

- For regression problems

$$h^*(x) = \mathbb{E}_{\theta}[y|x] \xrightarrow{\text{Plug in}} \hat{h}(x) = \mathbb{E}_{\hat{\theta}}[y|x]$$

- Linear Gaussian model: $y \sim N(x^T \hat{\theta}, \sigma^2)$

Today

- Naïve Bayes model
- Generalization error by bias-variance decomposition
 - Understand the problem of overfitting
- Learning risk decomposition
 - Understand machine learning

Recap: Prediction of a probabilistic model

- Step-by-step derivation:
 - $h(x) = \operatorname{argmax}_y P(y|x)$
 - $h(x) = \operatorname{argmax}_y \frac{P(y)P(x|y)}{P(x)}$
 - $h(x) = \operatorname{argmax}_y P(y)P(x|y)$
- Estimating $P(y)$ is easy, but how to estimate $P(x|y)$?

Recap: Direct modeling $P(x|y)$ is hard

Binary vectors, 2^3 rows +
binary output $Y \in \{0, 1\}$

x_1	x_2	x_3
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

1. At least 8 data points are needed to determine $p(x|y)$.

2. At least 2^d data points are needed if there are **d binary features**.

Maximum likelihood estimation (MLE) is one of the ways to model $P(x|y)$.

Naïve Bayes assumption

- Given the class label y , the features are conditional independent of each other.
 - $P(x|y) = \prod_j p(x_j|y)$
 - x_j is the j -th **feature**, not j -th data point!
- How to understand?
 - All features independently affect the label.

Model $p(x|y)$ with naïve Bayes assumption

Binary vectors, 2^3 rows +
binary output $Y \in \{0, 1\}$

x_1	x_2	x_3
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Discussion: How many data points are needed to determine $p(x|y)$ given binary output and d binary features?

$$2d$$

Why? When we use the first feature, we ignore all other features...

Naïve Bayes model

- Prediction of probabilistic model:
 - $h(x) = \operatorname{argmax}_y P(y|x)$
 - $h(x) = \operatorname{argmax}_y \frac{P(y)P(x|y)}{P(x)}$
 - $h(x) = \operatorname{argmax}_y P(y)P(x|y)$
 - $h_{\text{Naïve_Bayes}}(x) = \operatorname{argmax}_y P(y) \prod_{j=1}^d P(x_j|y)$
- Bayes rule
 - $P(y|x) = \frac{P(y)P(x|y)}{P(x)}$
- Naïve Bayes assumption
 - $P(x|y) = \prod_j p(x_j|y)$

In-class exercise: Naïve Bayes model for email filter

Email ID	Contains "buy"	Contains "cheap"	Contains "free"	Label
1	Yes	Yes	No	Spam
2	No	No	Yes	Not Spam
3	Yes	No	Yes	Spam
4	No	Yes	No	Not Spam
5	Yes	Yes	Yes	Spam
6	No	No	No	Not Spam

- Naïve Bayes model: $h_{\text{Naïve_Bayes}}(x) = \operatorname{argmax}_y P(y) \prod_{j=1}^d P(x_j|y)$
- Step 1: calculate $P(y)$
- Step 2: calculate $P(x_j|y)$
- Step 3: prediction on [Yes, No, No]

Naïve Bayes model with continuous variables

- So far we assumed a binomial or discrete distribution for the data given the model ($p(\mathbf{x}^i|y)$)
- However, in many cases the data contains continuous features:
 - Height, weight, Levels of genes in cells, Brain activity
- **Gaussian Naïve Bayes model:**

$$x_i|y \sim \mathcal{N}(\mu_y, \sigma_y^2)$$

Checkpoint: Probabilistic models

- Probabilistic / Generative / Bayesian models are very powerful and interpretable method for modeling the world.
 - Customize ML models for your applications
 - Explicitly model the dependence
- Key task: Modeling $P(x|y)$
 - Maximum likelihood estimation
 - Data are identically and independently distributed (i.i.d.)
 - Naïve Bayes model
 - Features are independent given the label

So far, we have learned a lot of ML algorithms

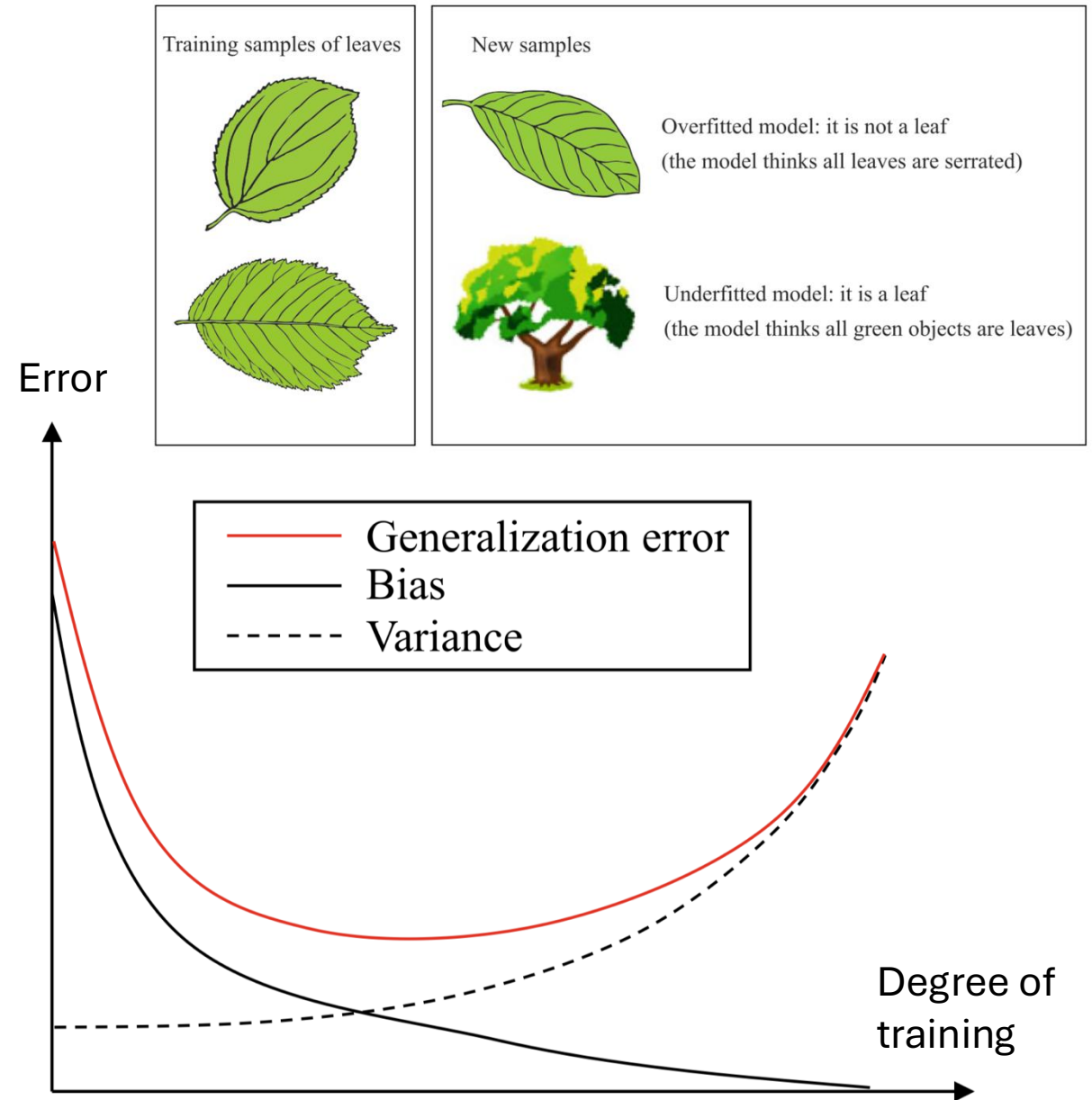
- The key goal of ML algorithms is to
 - Minimize the generalization error
 - We want to train a learning algorithm that works well on test data
- The ultimate goal of ML algorithms is to
 - Learn the best hypothesis!
- What are the factors that **systematically** affect learning process?

Bias-variance decomposition

- Definitions:
 - Feature: x
 - Label: $y = f(x) + \epsilon$
 - Label generating function: $f(x)$
 - Noise: $\epsilon, E[\epsilon] = 0, Var[\epsilon] = \sigma^2$
 - Prediction: $\hat{y}$
- Bias:
 - $|f(x) - E[\hat{y}]|$
- Variance:
 - $E[(\hat{y} - E[\hat{y}])^2]$
- Generalization error:
 - $E[(y - \hat{y})^2]$
- Generalization error decomposition:
 - $E[(y - \hat{y})^2] = Variance + bias^2 + \sigma^2$
 - Bias:
 - fitting of learning algorithm
 - Variance:
 - effect of given dataset
 - Noise:
 - difficulty of the learning problem

Bias-variance trade-off

- Generalization error decomposition:
 - $E[(y - \hat{y})^2] = \text{Variance} + \text{bias}^2 + \sigma^2$
- How to control the degree of training:
 - SGD: number of rounds
- Less training:
 - Model fitting is so weak, high bias
 - Underfitted
- Too much training:
 - Learned a lot of details of data, high variance
 - Overfitting



Loss, Empirical Risk, and Risk

- Loss function

$$\ell(h, (x, y))$$

- Empirical Risk function

$$\hat{R}(h, \text{Data}) = \frac{1}{n} \sum_{i=1}^n \ell(h, (x_i, y_i))$$

- (Population) Risk function

$$R(h, \mathcal{D}) = \mathbb{E}_{\mathcal{D}}[\ell(h, (x_i, y_i))]$$

Bayes optimal classifier, optimal classifier within the hypothesis class, Empirical Risk Minimizer

- Bayes Optimal classifier: $h_{\text{Bayes}} = \arg \min_h R(h)$
 - For 0-1 loss, the Bayes optimal classifier is

$$h_{\text{Bayes}} = \arg \max_y p(y|x) = \arg \max_y p(x|y)p(y)$$

- Optimal (within hypothesis class) classifier $h^* = \arg \min_{h \in \mathcal{H}} R(h)$
- ERM Classifier $h_{\text{ERM}} = \arg \min_{h \in \mathcal{H}} \hat{R}(h)$
- My classifier $\hat{h} = \text{My_Learning_Algorithm}(\text{Data})$

Risk Decomposition

$$\begin{aligned} & \mathbb{E}[R(\hat{h})] - R(h_{\text{Bayes}}) \\ \leq & \underbrace{\mathbb{E}[\hat{R}(\hat{h}) - \hat{R}(h_{\text{ERM}})]}_{\text{Optimization error}} + \underbrace{R(h^*) - R(h_{\text{Bayes}})}_{\text{Approximation error}} + \underbrace{\mathbb{E}[R(\hat{h}) - \hat{R}(\hat{h})]}_{\text{Generalization error}} \end{aligned}$$